

Elimination Rules Playbook

Evidence-based last-resort heuristics for Civil, Electrical, and Non-core MCQs.

Publication floor: each established rule has at least 15 independent supporting questions from FLTs, Subject Tests, and VALID PYQs. Target healthy band: 15–20+ examples.

How to apply these rules (exam workflow)

1. **Solve normally first** — formula, figure, code clause.
2. If stuck between two or three options, **classify the trap family** (geometry, pu base, sequence fault, service vs factored, etc.).
3. Open the matching rule below and run the **Application steps**.
4. **Strike** options that violate the rule; keep only survivors.
5. If two remain, use Rule 17 (units / figure consistency). Never use letter-position myths (pick C, longest option, middle value).

Drill practice: 50-question jumbled tests are in the same folder (Test 01 covers Rules 1–5, Test 02 Rules 6–10, Test 03 Rules 11–15, Test 04 Rules 16–17).

1. Established heuristics — index

#	Rule	Streams	Evidence
1	Diameter ↔ radius / area geometry slip	Civil, Electrical	129
2	Service vs factored / partial-factor mix	Civil	104
3	Gross vs net / hole deduction	Civil	95
4	Cover vs effective depth	Civil	40
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12	Rankine / Terzaghi / earth-pressure term drop	Civil	86
13	Mohr / principal-stress pair consistency	Civil	26
14	SFD↔BMD jump / couple vs point-load confusion	Civil	39

#	Rule	Streams	Evidence
15	SCR / PE latching vs holding vs firing	Electrical	18
16	Non-core: option that abandons the asked operation	Non-core	37
17	Last-resort: dimensional / absurdity cull (evidence-limited)	Civil, Electrical, Non-core	232

Stream coverage

- **Civil:** R1, R2, R3, R4, R10, R11, R12, R13, R14, R17
- **Electrical:** R1, R5, R6, R7, R8, R9, R10, R11, R15, R17
- **Non-core:** R16, R17

All three streams are covered by the established set.

2. Rules — explanation and application

Elimination Rule 1 — Diameter ↔ radius / area geometry slip

Evidence: ESTABLISHED — 129 independent (125 bank + 4 VALID PYQ) · **Streams:** Civil, Electrical

What this rule means

Circular geometry questions almost always hide a diameter-versus-radius mistake. Area is $A = \pi d^2/4$. Candidates who treat the given diameter as a radius get areas four times too large (or stresses four times too small).

When to use

Circular section, shaft, hanger, pipe, or bolt with diameter given in mm.

Application steps

1. Circle the numeric diameter in the stem and write $r = d/2$ on scratch paper.
2. Compute $A = \pi d^2/4$ once; do not reuse a remembered ' πr^2 ' with d plugged in as r .
3. Strike any option that matches $P/(\pi d^2)$, $P/(\pi r)$, or $4x/0.25x$ of the correct stress.
4. If net area of a plate with holes is asked, apply hole deductions after the correct gross area.

Memory cue: $A = \pi d^2/4$; $\sigma = P/A$

Core elimination move

Compute $A = \pi d^2/4$ (or $r = d/2$) once on scratch paper. Eliminate options that match $P/(\pi d^2)$, $P/(\pi r)$, or treating d as r .

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Civil FLT-01	data/civil/ce-flt01.js#Q1	Following a gantry-tie inspection at a substation, the engineer records a 20 mm diameter member under 55 kN...
Civil FLT-02	data/civil/ce-flt02.js#Q1	A 2.0 m long steel tie of uniform 500 mm ² area carries a gradually applied 50 kN service load. Neglecting s...
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q2	Water flows in a 0.2 m diameter pipe at mean velocity 4 m/s. Discharge is closest to:
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q1	An RCC beam of M25 and Fe500 has factored $M_u = 180$ kN m, $b = 230$ mm and $d = 450$ mm. Using $z = 0.9d$, the req...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q2	A 230 mm wide beam with effective depth 450 mm carries factored shear 125 kN at a section. Nominal shear st...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q4	A 300 mm × 300 mm short tied column carries factored axial load 1050 kN. The average factored stress P_u/Ag ...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q6	An RCC beam of M25 and Fe500 has factored $M_u = 185$ kN m, $b = 230$ mm and $d = 450$ mm. Using $z = 0.9d$, the req...

Source exam	Reference	Stem (truncated)
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q7	A 230 mm wide beam with effective depth 450 mm carries factored shear 130 kN at a section. Nominal shear st...

Elimination Rule 2 — Service vs factored / partial-factor mix

Evidence: ESTABLISHED — 104 independent (102 bank + 2 VALID PYQ) · **Streams:** Civil

What this rule means

Limit-state design mixes service loads, factored loads, and material strengths (f_y , f_{ck}). A classic trap is to treat f_y as the computed member stress, or to apply load factors twice.

When to use

Stem mixes service loads with limit-state options, or quotes f_y/f_{ck} beside M_u/V_u .

Application steps

1. Label the stem: SLS (service) or ULS (factored M_u/V_u).
2. If M_u/V_u is already factored, do not multiply by γ_f again.
3. Strike options that equal f_y or $0.87 f_y$ when the ask is a computed stress from P/A or M/Z .
4. Keep A_{st} calculations on the factored M_u and design stress $0.87 f_y$ (or the code path stated).

Memory cue: ULS actions use γ_f ; design steel stress is typically $0.87 f_y$ (IS 456 path)

Core elimination move

Ask: is this SLS or ULS? Strike options that apply γ_f to an already-factored action, or that treat f_y as the computed stress.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Civil FLT-01	data/civil/ce-flt01.js#Q1	Following a gantry-tie inspection at a substation, the engineer records a 20 mm diameter member under 55 kN...
Civil FLT-02	data/civil/ce-flt02.js#Q1	A 2.0 m long steel tie of uniform 500 mm ² area carries a gradually applied 50 kN service load. Neglecting s...
Civil ST-FE	data/civil/st/ce-st-fe-01.js#Q3	The shown square footing distributes a concentric service load of 260 kN over a 2.3 m × 2.3 m base. The ave...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q1	An RCC beam of M25 and Fe500 has factored $M_u = 180$ kN m, $b = 230$ mm and $d = 450$ mm. Using $z = 0.9d$, the req...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q2	A 230 mm wide beam with effective depth 450 mm carries factored shear 125 kN at a section. Nominal shear st...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q4	A 300 mm × 300 mm short tied column carries factored axial load 1050 kN. The average factored stress P_u/A_g ...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q5	A square footing 2.1 m by 2.1 m carries a service column load of 190 kN (self-weight ignored). Average soil...

Source exam	Reference	Stem (truncated)
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q6	An RCC beam of M25 and Fe500 has factored $M_u = 185 \text{ kN m}$, $b = 230 \text{ mm}$ and $d = 450 \text{ mm}$. Using $z = 0.9d$, the req...

Elimination Rule 3 — Gross vs net / hole deduction

Evidence: ESTABLISHED — 95 independent (95 bank + 0 VALID PYQ) · **Streams:** Civil

What this rule means

Tension capacity of bolted plates uses net area, not gross. For staggered holes, add $s^2/(4g)$ terms. Options that equal gross area ignore the holes entirely.

When to use

Tension member, plate with bolt holes, staggered pitch, or chain of holes.

Application steps

1. Write $A_{net} = (\text{gross width} - n \cdot d_{\text{hole}}) \cdot t$ (+ stagger terms if shown).
2. Strike the gross-area option immediately.
3. If stagger is drawn, include $s^2/(4g)$; if chain is transverse only, do not invent stagger credit.

Memory cue: $A_{net} \sim (b - n \cdot d_h) \cdot t$; staggered: + $\sum s^2/(4g)$

Core elimination move

Net = gross – holes ($\pm s^2/4g$). Eliminate options that equal gross area, or that deduct diameter without adding the staggered term when stagger is shown.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Civil FLT-01	data/civil/ce-flt01.js#Q1	Following a gantry-tie inspection at a substation, the engineer records a 20 mm diameter member under 55 kN...
Civil FLT-02	data/civil/ce-flt02.js#Q20	Two short tied columns have the same gross area and longitudinal-steel percentage. Column X alone receives ...
Civil ST-FE	data/civil/st/ce-st-fe-01.js#Q1	For a strip footing, take $c = 20$ kPa, $N_c = 25$, $q = 30$ kPa, $N_q = 12$, $\gamma = 18$ kN/m ³ , $B = 2$ m and $N_y = 8$. Terza...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q4	A 300 mm × 300 mm short tied column carries factored axial load 1050 kN. The average factored stress P_u/Ag ...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q9	A 300 mm × 300 mm short tied column carries factored axial load 900 kN. The average factored stress P_u/Ag i...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q24	A 300 mm × 300 mm short tied column carries factored axial load 1050 kN. The average factored stress P_u/Ag ...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q29	A 300 mm × 300 mm short tied column carries factored axial load 900 kN. The average factored stress P_u/Ag i...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q34	A 300 mm × 300 mm short tied column carries factored axial load 950 kN. The average factored stress P_u/Ag i...

Elimination Rule 4 — Cover vs effective depth

Evidence: ESTABLISHED — 40 independent (39 bank + 1 VALID PYQ) · **Streams:** Civil

What this rule means

RCC flexural and shear formulas use effective depth d , not overall depth D . Cover and bar diameter set $d = D - \text{cover} - \phi/2$ (as defined in the stem).

When to use

RCC beam/slab with overall D , cover, bar dia — options in A_{st} , τ_v , or lever arm.

Application steps

1. Extract D , cover, and bar size; compute d before A_{st} or τ_v .
2. Strike options that use D in place of d in $M_u = 0.87 f_y A_{st} z$ or $\tau_v = V_u/(b d)$.

Memory cue: $d = D - \text{cover} - \phi/2$; $\tau_v = V_u/(b d)$

Core elimination move

$d = D - \text{cover} - \phi/2$ (or as stem defines). Strike options that use overall depth as effective depth.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Civil FLT-01	data/civil/ce-flt01.js#Q16	Before a pour on an RCC beam in moderate exposure, measured cover is below the durability requirement altho...
Civil FLT-02	data/civil/ce-flt02.js#Q16	An RCC beam in moderate exposure is detailed with nominal cover smaller than the durability requirement, th...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q1	An RCC beam of M25 and Fe500 has factored $M_u = 180$ kN m, $b = 230$ mm and $d = 450$ mm. Using $z = 0.9d$, the req...
Civil ST-STEEL	data/civil/st/ce-st-steel-01.js#Q4	A lap connection has 4 bearing-type bolts, each with governing design shear capacity 48 kN. Ignoring eccent...
Civil ST-STEEL	data/civil/st/ce-st-steel-01.js#Q9	A lap connection has 4 bearing-type bolts, each with governing design shear capacity 45 kN. Ignoring eccent...
Civil ST-STEEL	data/civil/st/ce-st-steel-01.js#Q24	A lap connection has 4 bearing-type bolts, each with governing design shear capacity 48 kN. Ignoring eccent...
Civil ST-STEEL	data/civil/st/ce-st-steel-01.js#Q29	A lap connection has 4 bearing-type bolts, each with governing design shear capacity 45 kN. Ignoring eccent...
Civil ST-STEEL	data/civil/st/ce-st-steel-01.js#Q34	A lap connection has 4 bearing-type bolts, each with governing design shear capacity 46 kN. Ignoring eccent...

Elimination Rule 5 — Per-unit base-change scaling

Evidence: ESTABLISHED — 95 independent (95 bank + 0 VALID PYQ) · **Streams:** Electrical

What this rule means

Per-unit impedance scales with the chosen bases: $Z_{pu} = Z_{ohm} * S_{base} / V_{base}^2$. Changing only S_{base} scales Z_{pu} in direct proportion.

When to use

Z_{pu} / X_{pu} and base MVA or kV changes.

Application steps

1. Note old and new S_{base} and V_{base} .
2. If S_{base} doubles and V_{base} is unchanged, Z_{pu} halves — strike 'unchanged', 'doubles', 'squared'.
3. If V_{base} also changes, recompute the full ratio.

Memory cue: $Z_{pu} = Z_{ohm} * S_{base} / V_{base}^2$

Core elimination move

$Z_{pu} \propto S_{base} / V_{base}^2$. If S_{base} doubles and V_{base} fixed → Z_{pu} halves. Eliminate 'unchanged', 'doubles', or 'squared' unless the stem changed V_{base} too.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Electrical FLT-01	data/electrical/ee-flt01.js#Q1	A lossless 400 kV line has $Z_c = 250 \Omega$. Its SIL (MW) is closest to:
Electrical FLT-02	data/electrical/ee-flt02.js#Q2	On 200 MVA, 400 kV base, a series reactor is 96 ohms. Per-unit reactance is closest to:
Electrical ST-PS	data/electrical/st/ee-st-ps-01.js#Q1	On a common 100 MVA base, if the MVA base is doubled while the base kV is unchanged, a line reactance expre...
Electrical FLT-01	data/electrical/ee-flt01.js#Q2	On 100 MVA, 220 kV base, a line reactance 48.4Ω equals how many pu?
Electrical FLT-01	data/electrical/ee-flt01.js#Q3	A 3- ϕ feeder has positive-sequence impedance $j0.2$ pu and zero-sequence $j0.5$ pu on same base. For a solid SL...
Electrical FLT-01	data/electrical/ee-flt01.js#Q5	Two identical 50 MVA, 11 kV, $X_d''=0.15$ pu machines feed a bus. On 100 MVA base, the bus short-circuit MVA fo...
Electrical FLT-01	data/electrical/ee-flt01.js#Q6	Ferranti rise on a long EHV line is most severe when:
Electrical FLT-01	data/electrical/ee-flt01.js#Q9	From the open-conductor / sequence figure shown (use only labelled phase currents; $I_c = 0$), the zero-sequen...

Elimination Rule 6 — Sequence-network connection by fault type

Evidence: ESTABLISHED — 25 independent (18 bank + 7 VALID PYQ) · **Streams:** Electrical

What this rule means

Unsymmetrical faults require specific sequence-network connections. Memorising the four classic connections eliminates most wrong options instantly.

When to use

LG / LL / LLG / 3 ϕ fault; options name sequence networks.

Application steps

1. Identify fault type: 3-ph, LG, LL, LLG.
2. 3-ph $\rightarrow$ positive only; LG $\rightarrow$ 1-2-0 in series; LL $\rightarrow$ 1 parallel 2; LLG $\rightarrow$ 1 series (2 parallel 0).
3. Strike any option that omits a required sequence or uses only zero-sequence.

Memory cue: LG: $Z_1+Z_2+Z_0$; LL: $Z_1 || Z_2$; LLG: $Z_1+(Z_2 || Z_0)$

Core elimination move

3 ϕ $\rightarrow$ positive only. LG $\rightarrow$ series 1-2-0. LL $\rightarrow$ 1 parallel 2. LLG $\rightarrow$ 1 series (2||0). Eliminate any option that omits a required sequence or uses only zero-sequence.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Electrical FLT-01	data/electrical/ee-flt01.js#Q3	A 3- ϕ feeder has positive-sequence impedance j0.2 pu and zero-sequence j0.5 pu on same base. For a solid SL...
Electrical FLT-02	data/electrical/ee-flt02.js#Q3	Solid LG fault: $Z_1=Z_2=j0.15$ pu, $Z_0=j0.60$ pu, $V_{pref}=1$ pu. $ I_f $ (pu) nearest:
Electrical ST-EM	data/electrical/st/ee-st-em-01.js#Q17	A synchronous generator is being synchronised with live busbars. Which statement is correct?
Electrical ST-PS	data/electrical/st/ee-st-ps-01.js#Q2	For a double line-to-ground fault on an unloaded generator, the sequence networks required for the fault ca...
Electrical ST-PS	data/electrical/st/ee-st-ps-01.js#Q12	Zero-sequence network of grounded-star alternator through X_n uses:
Electrical ST-PS	data/electrical/st/ee-st-ps-01.js#Q24	An unsymmetrical line-to-line-to-ground fault is being analysed by symmetrical components. Which statement ...
Electrical ST-PS	data/electrical/st/ee-st-ps-01.js#Q34	An unsymmetrical line-to-line-to-ground fault is being analysed by symmetrical components. Which statement ...
Electrical ST-PS	data/electrical/st/ee-st-ps-01.js#Q44	An unsymmetrical line-to-line-to-ground fault is being analysed by symmetrical components. Which statement ...

Elimination Rule 7 — $\sqrt{3}$ / phase-factor forgotten

Evidence: ESTABLISHED — 24 independent (18 bank + 6 VALID PYQ) · **Streams:** Electrical

What this rule means

Three-phase power and current conversions carry a root-3 factor. Forgetting it, or using 3 instead of root-3, produces the most common distractors.

When to use

3-phase power, line vs phase quantities, line current from kVA.

Application steps

1. Write $P = \sqrt{3} \cdot V_L \cdot I_L \cdot \cos(\phi)$ (or the form asked).
2. Strike options that drop root3 or replace it with 3 without justification.

Memory cue: $P = \sqrt{3} \cdot V_L \cdot I_L \cdot \cos\phi$

Core elimination move

$P = \sqrt{3} V_L I_L \cos\phi$. Eliminate options that drop $\sqrt{3}$ or use 3 instead of $\sqrt{3}$ (or vice versa) without stem justification.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Electrical FLT-01	data/electrical/ee-flt01.js#Q3	A 3- ϕ feeder has positive-sequence impedance j0.2 pu and zero-sequence j0.5 pu on same base. For a solid SL...
Electrical FLT-02	data/electrical/ee-flt02.js#Q8	During a bolted three-phase short circuit on an unloaded radial feeder, which sequence network combination ...
Electrical ST-EC	data/electrical/st/ee-st-ec-01.js#Q17	A balanced three-phase load is measured with the two-wattmeter method. Which statement is correct?
Electrical ST-MEAS	data/electrical/st/ee-st-meas-01.js#Q8	Two-wattmeter method pf=:
Electrical ST-PE	data/electrical/st/ee-st-pe-01.js#Q9	180° mode three-phase VSI line voltage steps:
Electrical ST-PS	data/electrical/st/ee-st-ps-01.js#Q3	Following a severe three-phase disturbance in a single-machine infinite-bus system, the equal-area criterio...
VALID PYQ · 2022EE.pdf	PYQ:2022EE.pdf#Q21	In the circuit shown below, a three-phase star-connected unbalanced load is connected to a balanced three-p...
VALID PYQ · 2023EE.pdf	PYQ:2023EE.pdf#Q39	A 3-phase, star-connected, balanced load is supplied from a 3-phase, 400 V (rms), balanced voltage source w...

Elimination Rule 8 — Thevenin / Norton source deactivation

Evidence: ESTABLISHED — 18 independent (14 bank + 4 VALID PYQ) · **Streams:** Electrical

What this rule means

Thevenin/Norton resistance is found with independent sources deactivated: voltage sources shorted, current sources opened.

When to use

Req, RN, or Vth with independent sources in the figure.

Application steps

1. For Req/RN, deactivate independent sources first.
2. Strike options that leave sources 'on' while claiming equivalent resistance.
3. Do not confuse Vth (open-circuit voltage) with RN.

Memory cue: Independent V → short; independent I → open; then find Req

Core elimination move

Independent voltage sources → short; current sources → open. Eliminate options that leave sources active while claiming Req/RN.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Electrical FLT-01	data/electrical/ee-flt01.js#Q29	In the resistor network shown driven by Vs, the equivalent resistance Req looking into the right-hand termi...
Electrical FLT-02	data/electrical/ee-flt02.js#Q35	Superposition theorem applies to:
Electrical ST-EC	data/electrical/st/ee-st-ec-01.js#Q3	After independent sources are deactivated, the Norton resistance seen at the output terminals is:
VALID PYQ · 2021EE.pdf	PYQ:2021EE.pdf#Q4	For the network shown, the equivalent Thevenin voltage and Thevenin impedance as seen across terminals 'ab' is
VALID PYQ · 2026EE.pdf	PYQ:2026EE.pdf#Q24	For the circuit shown, which one of the following options correctly identifies the Thevenin's equivalent pa...
VALID PYQ · 2026EE.pdf	PYQ:2026EE.pdf#Q44	The terminal voltage and current of a linear electrical network shown in Figure
VALID PYQ · AP-GENCO-Tech-2012.pdf	PYQ:AP-GENCO-Tech-2012.pdf#Q56	The Norton's resistance of the circuit shown is
Electrical FLT-01	data/electrical/ee-flt01.js#Q34	An ideal ammeter is connected between terminals A and B in the circuit shown. The ammeter reading is:

Elimination Rule 9 — Resonance / $X_L = X_C$ identity

Evidence: ESTABLISHED — 47 independent (44 bank + 3 VALID PYQ) · **Streams:** Electrical

What this rule means

At series resonance $X_L = X_C$ so net reactive part is zero and $Z = R$ (minimum). Parallel resonance is the dual (impedance maximum).

When to use

Series/parallel RLC, ω_0 , or impedance at resonance.

Application steps

1. Confirm series vs parallel from the figure/stem.
2. At series resonance strike any option that still carries $j(X_L - X_C) \neq 0$.
3. Use $\omega_0 = 1/\sqrt{LC}$ only when that is the ask.

Memory cue: Series resonance: $X_L = X_C \rightarrow Z = R$

Core elimination move

Series resonance $\rightarrow Z = R$ (min). Parallel $\rightarrow Z$ max. Eliminate options that keep $j(X_L - X_C)$ nonzero at the stated resonance.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Electrical FLT-01	data/electrical/ee-flt01.js#Q13	In equal-area criterion, for a sudden load rejection on a machine, accelerating area is bounded by:
Electrical FLT-02	data/electrical/ee-flt02.js#Q18	From the alternator phasor diagram (I_a leading V_t), the machine is operating as:
Electrical ST-EC	data/electrical/st/ee-st-ec-01.js#Q11	Source-free series RLC underdamped $\omega_d =$:
Electrical ST-EM	data/electrical/st/ee-st-em-01.js#Q3	Sync machine: overexcited on infinite bus behaves as:
Electrical ST-EM	data/electrical/st/ee-st-em-01.js#Q14	A synchronous generator is operated at fixed terminal voltage and real power while excitation is varied. Wh...
Electrical ST-EM	data/electrical/st/ee-st-em-01.js#Q20	A synchronous generator is operated at fixed terminal voltage and real power while excitation is varied. Wh...
Electrical ST-EM	data/electrical/st/ee-st-em-01.js#Q26	A synchronous generator is operated at fixed terminal voltage and real power while excitation is varied. Wh...
Electrical ST-EM	data/electrical/st/ee-st-em-01.js#Q32	A synchronous generator is operated at fixed terminal voltage and real power while excitation is varied. Wh...

Elimination Rule 10 — Sign / sense / tension–compression flip

Evidence: ESTABLISHED — 270 independent (259 bank + 11 VALID PYQ) · **Streams:** Civil, Electrical

What this rule means

Many options are numerically close but reverse tension/compression or reaction direction. Equilibrium sense is part of the answer.

When to use

Reactions, BMD jump, member force T/C, or direction in options.

Application steps

1. Fix one free-body sign convention from the figure.
2. Eliminate options with correct magnitude but wrong sense (T/C or up/down).

Memory cue: $\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$ (sense matters)

Core elimination move

Fix one free-body sign convention from the figure. Eliminate options whose magnitude is right but sense ($\uparrow/\downarrow$, T/C) contradicts equilibrium.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Civil FLT-01	data/civil/ce-flt01.js#Q1	Following a gantry-tie inspection at a substation, the engineer records a 20 mm diameter member under 55 kN...
Civil FLT-02	data/civil/ce-flt02.js#Q1	A 2.0 m long steel tie of uniform 500 mm ² area carries a gradually applied 50 kN service load. Neglecting s...
Civil ST-FE	data/civil/st/ce-st-fe-01.js#Q4	A pile group contains 4 piles. Each pile has ultimate base resistance 600 kPa over toe area 0.2 m ² ; shaft r...
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q2	Water flows in a 0.2 m diameter pipe at mean velocity 4 m/s. Discharge is closest to:
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q3	The shown venturimeter is installed in a 0.2 m diameter inlet pipe. If the inlet velocity is 3 m/s, the dis...
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q4	A horizontal jet of discharge 0.1 m ³ /s accelerates from 5 m/s to 15 m/s in a vane. Neglecting pressure forc...
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q7	Water flows in a 0.2 m diameter pipe at mean velocity 3 m/s. Discharge is closest to:
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q8	The shown venturimeter is installed in a 0.2 m diameter inlet pipe. If the inlet velocity is 5 m/s, the dis...

Elimination Rule 11 — Figure-dependency: cover the figure test

Evidence: ESTABLISHED — 304 independent (109 bank + 195 VALID PYQ) · **Streams:** Civil, Electrical

What this rule means

If a figure is attached, the keyed answer usually depends on a labelled length, support, or curve. Solving from the stem alone is a dependency failure.

When to use

Stem says 'shown', 'figure', or options name Detail/Curve labels.

Application steps

1. Read every dimension and support on the crop before picking.
2. Strike options that ignore a labelled overhang, load, or curve name.
3. If the stem names Detail/Curve labels, those labels must appear on the figure.

Memory cue: Cover-the-figure test: cannot answer correctly with figure hidden

Core elimination move

If you can pick the answer without the figure, stop — stem/figure mismatch. Otherwise eliminate options that ignore a labelled dimension, support, or curve on the crop.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Civil FLT-01	data/civil/ce-flt01.js#Q3	Proof-load planning uses an overhanging beam A–B–C with supports at A and B: AB = 4 m, overhang BC = 2 m, U...
Civil FLT-02	data/civil/ce-flt02.js#Q3	The overhanging beam in the crop carries the shown point load and UDL; span lengths and load locations appe...
Civil ST-FE	data/civil/st/ce-st-fe-01.js#Q3	The shown square footing distributes a concentric service load of 260 kN over a 2.3 m × 2.3 m base. The ave...
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q3	The shown venturimeter is installed in a 0.2 m diameter inlet pipe. If the inlet velocity is 3 m/s, the dis...
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q8	The shown venturimeter is installed in a 0.2 m diameter inlet pipe. If the inlet velocity is 5 m/s, the dis...
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q14	The shown venturimeter is installed in a 0.2 m diameter inlet pipe. If the inlet velocity is 5 m/s, the dis...
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q19	The shown venturimeter is installed in a 0.2 m diameter inlet pipe. If the inlet velocity is 4 m/s, the dis...
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q25	The shown venturimeter is installed in a 0.2 m diameter inlet pipe. If the inlet velocity is 4 m/s, the dis...

Elimination Rule 12 — Rankine / Terzaghi / earth-pressure term drop

Evidence: ESTABLISHED — 86 independent (85 bank + 1 VALID PYQ) · **Streams:** Civil

What this rule means

Bearing capacity and earth-pressure formulae have multiple additive terms. Dropping N_y , swapping K_a/K_p , or using dry unit weight under water are classic traps.

When to use

Bearing capacity or lateral earth pressure with N_c , N_q , N_y or K_a/K_p .

Application steps

1. Write the full Terzaghi or Rankine expression used in the stem.
2. Strike options that omit a term or use K_a where K_p is required (or vice versa).
3. If submerged, use buoyant unit weight as required.

Memory cue: $q_u = c N_c + q N_q + 0.5 \gamma B N_y$ (Terzaghi form)

Core elimination move

Write the full Terzaghi/Rankine expression. Eliminate options that drop the $\gamma B N_y$ term, swap $K_a \leftrightarrow K_p$, or use dry γ when stem says submerged.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Civil FLT-01	data/civil/ce-flt01.js#Q11	During buckling checks for a slender pin-ended strut, a junior engineer proposes to use the classical Euler...
Civil FLT-02	data/civil/ce-flt02.js#Q9	A simply supported deep timber beam develops horizontal cracks near the neutral axis although bending stres...
Civil ST-FE	data/civil/st/ce-st-fe-01.js#Q1	For a strip footing, take $c = 20$ kPa, $N_c = 25$, $q = 30$ kPa, $N_q = 12$, $\gamma = 18$ kN/m ³ , $B = 2$ m and $N_y = 8$. Terza...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q5	A square footing 2.1 m by 2.1 m carries a service column load of 190 kN (self-weight ignored). Average soil...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q10	A square footing 2 m by 2 m carries a service column load of 180 kN (self-weight ignored). Average soil pre...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q15	A square footing 2.2 m by 2.2 m carries a service column load of 200 kN (self-weight ignored). Average soil...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q20	A square footing 2.1 m by 2.1 m carries a service column load of 190 kN (self-weight ignored). Average soil...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q30	A square footing 2.2 m by 2.2 m carries a service column load of 200 kN (self-weight ignored). Average soil...

Elimination Rule 13 — Mohr / principal-stress pair consistency

Evidence: ESTABLISHED — 26 independent (23 bank + 3 VALID PYQ) · **Streams:** Civil

What this rule means

Principal stresses from Mohr's circle must satisfy $\sigma_1 + \sigma_2 = \sigma_x + \sigma_y$. Treating shear as a normal stress breaks that invariant.

When to use

σ_x , σ_y , τ_{xy} given; options are σ_1 , σ_2 , or θ_p .

Application steps

1. Compute σ_{avg} and R ; then $\sigma_{1,2} = \sigma_{avg} \pm R$.
2. Strike pairs that violate $\sigma_1 + \sigma_2 = \sigma_x + \sigma_y$.

Memory cue: $\sigma_{1,2} = (\sigma_x + \sigma_y)/2 \pm \sqrt{[(\sigma_x - \sigma_y)/2]^2 + \tau^2}$

Core elimination move

$\sigma_{avg} = (\sigma_x + \sigma_y)/2$; $R = \sqrt{[(\sigma_x - \sigma_y)/2]^2 + \tau^2}$. Eliminate pairs that violate $\sigma_1 + \sigma_2 = \sigma_x + \sigma_y$ or that treat τ as a normal stress.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Civil FLT-01	data/civil/ce-flt01.js#Q5	While interpreting a strain-rosette investigation on a critical plane-stress element, the engineer reviews ...
Civil FLT-02	data/civil/ce-flt02.js#Q12	At a point, $\sigma_x = 80$ MPa tension, $\sigma_y = 20$ MPa compression and $\tau_{xy} = 30$ MPa. Determin...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q4	A 300 mm × 300 mm short tied column carries factored axial load 1050 kN. The average factored stress P_u/A_g ...
Civil ST-SOM	data/civil/st/ce-st-som-01.js#Q13	At a point in a plate, $\sigma_x = 92$ MPa, $\sigma_y = 22$ MPa and $\tau_{xy} = 30$ MPa. The major principal stress is closest to:
Civil ST-SOM	data/civil/st/ce-st-som-01.js#Q18	At a point in a plate, $\sigma_x = 97$ MPa, $\sigma_y = 22$ MPa and $\tau_{xy} = 35$ MPa. The major principal stress is closest to:
Civil ST-SOM	data/civil/st/ce-st-som-01.js#Q23	At a point in a plate, $\sigma_x = 102$ MPa, $\sigma_y = 22$ MPa and $\tau_{xy} = 34$ MPa. The major principal stress is closest to:
Civil ST-SOM	data/civil/st/ce-st-som-01.js#Q28	At a point in a plate, $\sigma_x = 107$ MPa, $\sigma_y = 22$ MPa and $\tau_{xy} = 33$ MPa. The major principal stress is closest to:
Civil ST-SOM	data/civil/st/ce-st-som-01.js#Q33	At a point in a plate, $\sigma_x = 112$ MPa, $\sigma_y = 22$ MPa and $\tau_{xy} = 32$ MPa. The major principal stress is closest to:

Elimination Rule 14 — SFD↔BMD jump / couple vs point-load confusion

Evidence: ESTABLISHED — 39 independent (34 bank + 5 VALID PYQ) · **Streams:** Civil

What this rule means

On SFDs/BMDs: a point load causes a shear jump; a couple causes a moment jump (no shear jump).

When to use

Beam figure with point load, UDL, or couple; options describe jumps.

Application steps

1. Identify whether the singularity is a force or a couple.
2. Strike options that swap shear-jump and moment-jump rules.

Memory cue: Point load → ΔV ; Couple → ΔM

Core elimination move

Point load → shear jump; couple → moment jump (no shear jump). Eliminate options that swap those rules.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Civil FLT-01	data/civil/ce-flt01.js#Q9	On the plotted bending-moment diagram, a marked vertical jump occurs at section C while shear remains finit...
Civil FLT-02	data/civil/ce-flt02.js#Q14	Match loading change with the qualitative SFD/BMD consequence: (1) point load, (2) applied couple, (3) UDL,...
Civil ST-RCC	data/civil/st/ce-st-rcc-01.js#Q1	An RCC beam of M25 and Fe500 has factored $M_u = 180$ kN m, $b = 230$ mm and $d = 450$ mm. Using $z = 0.9d$, the req...
Civil ST-SOM	data/civil/st/ce-st-som-01.js#Q2	A simply supported beam of span 5 m carries a uniform load of 13 kN/m over the full span. The maximum bendi...
Civil ST-SOM	data/civil/st/ce-st-som-01.js#Q7	A simply supported beam of span 4 m carries a uniform load of 14 kN/m over the full span. The maximum bendi...
Civil ST-SOM	data/civil/st/ce-st-som-01.js#Q12	A simply supported beam of span 6 m carries a uniform load of 15 kN/m over the full span. The maximum bendi...
Civil ST-SOM	data/civil/st/ce-st-som-01.js#Q17	A simply supported beam of span 5 m carries a uniform load of 12 kN/m over the full span. The maximum bendi...
Civil ST-SOM	data/civil/st/ce-st-som-01.js#Q22	A simply supported beam of span 4 m carries a uniform load of 13 kN/m over the full span. The maximum bendi...

Elimination Rule 15 — SCR / PE latching vs holding vs firing

Evidence: ESTABLISHED — 18 independent (14 bank + 4 VALID PYQ) · **Streams:** Electrical

What this rule means

For SCRs, latching current is larger than holding current. Latching is needed to establish conduction after the gate pulse; holding keeps it on.

When to use

Thyristor/SCR gate, latching, holding, or commutation options.

Application steps

1. Remember $I_L > I_H$.
2. Strike options that swap latching with holding or that size gate width from holding alone.

Memory cue: Latching current > holding current

Core elimination move

Latching > holding; latching is the current to *establish* conduction after gate pulse. Eliminate options that swap latching ↔ holding or that set gate width from holding current alone.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Electrical FLT-01	data/electrical/ee-flt01.js#Q47	An SCR turns off when which condition is met for sufficient time?
Electrical FLT-02	data/electrical/ee-flt02.js#Q47	After latch, SCR stays on until anode current falls below:
Electrical ST-PE	data/electrical/st/ee-st-pe-01.js#Q1	SCR latching current is:
VALID PYQ · 2026EE.pdf	PYQ:2026EE.pdf#Q28	Consider the circuit shown in Figure
VALID PYQ · APEPDCL-Assistant-Engineer-2019-Official-Paper.pdf	PYQ:APEPDCL-Assistant-Engineer-2019-Official-Paper.pdf#Q77	For the converter shown below, firing angle for thyristors 'T1' and 'T2' is $\pi/6$; firing angle for thyrist...
VALID PYQ · APEPDCL-Assistant-Engineer-2019-Official-Paper.pdf	PYQ:APEPDCL-Assistant-Engineer-2019-Official-Paper.pdf#Q78	In the figure shown below, a thyristor-based 1- ϕ voltage regulator with firing angle $\alpha = 90^\circ$. The rms out...
VALID PYQ · AP-GENCO-Tech-2012.pdf	PYQ:AP-GENCO-Tech-2012.pdf#Q10	The latching current of T1 is 1 mA. The minimum width of gate pulse required to turn on SCR is
Electrical FLT-02	data/electrical/ee-flt02.js#Q53	In field, class-F (line) commutation of SCR succeeds when:

Elimination Rule 16 — Non-core: option that abandons the asked operation

Evidence: ESTABLISHED — 37 independent (37 bank + 0 VALID PYQ) · **Streams:** Non-core

What this rule means

Non-core items often include one option that solves a different operation (CI instead of SI, active instead of passive, wrong percentage base).

When to use

Quant / reasoning / English — one option changes the operation (% , SI, tense, code).

Application steps

1. Re-state the exact ask in one short sentence.
2. Strike any option that changes the operation.

Memory cue: Match the asked operation exactly (% , SI, tense, code rule)

Core elimination move

Re-state the exact ask (20% of 450; SI for 2 years; passive of present perfect). Eliminate options that solve a different operation (CI instead of SI, active instead of passive).

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Civil FLT-01	data/civil/ce-flt01.js#Q72	A transmission-estimate package rises from ₹8.0 crore to ₹9.2 crore after a design revision. The percentage...
Civil FLT-02	data/civil/ce-flt02.js#Q73	A price rises 20%, falls 10%, then attracts 5% tax. Relative to the original pre-tax price, the final amoun...
Electrical FLT-01	data/electrical/ee-flt01.js#Q72	A transmission-estimate package rises from ₹8.0 crore to ₹9.2 crore after a design revision. The percentage...
Electrical FLT-02	data/electrical/ee-flt02.js#Q77	Contractor buys equipment for Rs 240000 and sells for Rs 276000. Profit percent on cost is:
Electrical FLT-02	data/electrical/ee-flt02.js#Q92	Choose passive: The technician calibrated the protection relay.:
Non-core ST-AR	data/noncore/st/nc-st-ar-01.js#Q1	A stores ledger records 450 cable drums. The value of 20% of this quantity is:
Non-core ST-AR	data/noncore/st/nc-st-ar-01.js#Q2	Simple interest on ₹10,000 at 10% per annum for 2 years is:
Non-core ST-AR	data/noncore/st/nc-st-ar-01.js#Q3	Average of 5,7,9,11,13:

Elimination Rule 17 — Last-resort: dimensional / absurdity cull (evidence-limited)

Evidence: ESTABLISHED — 232 independent (231 bank + 1 VALID PYQ) · **Streams:** Civil, Electrical, Non-core

What this rule means

When two options remain, cull any survivor whose units cannot equal the asked quantity (mm^2 vs N/mm^2 , ohm vs siemens, kN vs kN·m).

When to use

Time pressure; two options already killed by other established rules; two remain.

Application steps

1. Write the unit of the asked quantity.
2. Strike options whose units cannot match.
3. Then prefer consistency with figure boundary conditions — never 'pick C' myths.

Memory cue: Units(asked) must equal Units(option)

Core elimination move

Cull any remaining option whose units cannot match the ask (mm^2 vs N/mm^2 , Ω vs S, kN vs kN·m).

Then prefer the option consistent with the figure's boundary conditions. Do **not** guess from 'middle value' myths — our banks do not support that heuristic.

Evidence examples (sample)

Source exam	Reference	Stem (truncated)
Civil FLT-01	data/civil/ce-flt01.js#Q41	The plane truss shown carries load W and moment M at apex B. Force in member AB is:
Civil FLT-02	data/civil/ce-flt02.js#Q47	The design buckling-curve crop plots reduction factor χ against nondimensional slenderness for curve...
Civil ST-FE	data/civil/st/ce-st-fe-01.js#Q1	For a strip footing, take $c = 20 \text{ kPa}$, $N_c = 25$, $q = 30 \text{ kPa}$, $N_q = 12$, $\gamma = 18 \text{ kN}/\text{m}^3$, $B = 2 \text{ m}$ and $N_\gamma = 8$. Terza...
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q1	A mercury manometer connected to water shows a level difference of 0.4 m. Taking mercury specific gravity a...
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q2	Water flows in a 0.2 m diameter pipe at mean velocity 4 m/s. Discharge is closest to:
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q3	The shown venturimeter is installed in a 0.2 m diameter inlet pipe. If the inlet velocity is 3 m/s, the dis...
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q5	A centrifugal pump delivers $0.08 \text{ m}^3/\text{s}$ against 35 m head at overall efficiency 0.75. Water power input conve...
Civil ST-FM	data/civil/st/ce-st-fm-01.js#Q6	A mercury manometer connected to water shows a level difference of 0.5 m. Taking mercury specific gravity a...

3. Draft rules (not established — do not teach as heuristics)

These candidates have fewer than 15 independent hits. Keep them as a watchlist only.

- Kill the neighbour-formula lookalike — 4 hits
- SIL / surge-impedance identity — 5 hits
- Transformer referral / turns-ratio scaling — 7 hits

4. Anti-rules (unsupported myths)

- **Pick C when unsure** — keys are balanced; letter myths fail.
- **Longest option is correct** — distractors are equal-length engineering mistakes.
- **Always middle numerical value** — middle is often the classic slip ($\times 2$ / $\div 2$).
- **Ignore the figure if the stem looks complete** — imaged items are figure-dependent.

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